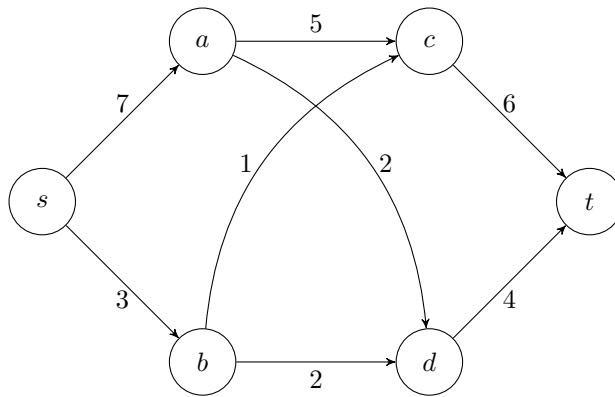


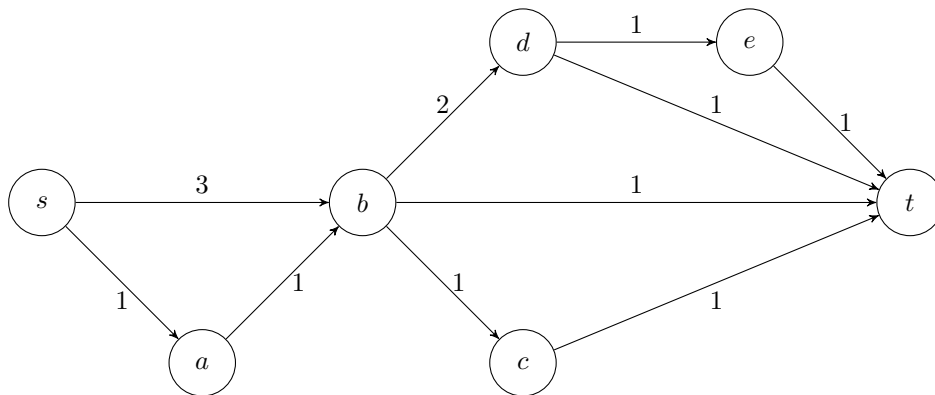
# Weekly Assignment 7: Flow algorithms

October 2025

1. (15 points) Run the capacity scaling algorithm on the following flow network.

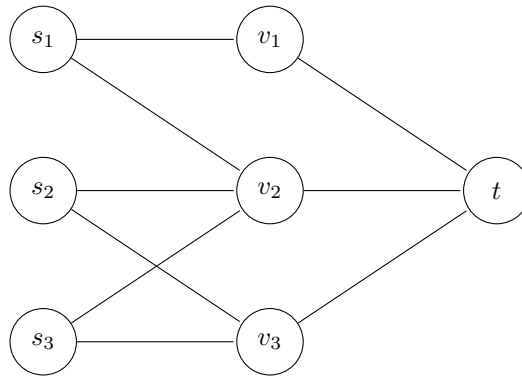


2. (15 points) Run the Edmonds-Karp algorithm on the following flow network.



3. (25 points) Suppose, we have a graph  $G = (V, E)$ , a subset  $S \subseteq V$  of source vertices, and a target vertex  $t \in V$ . A collection of **edge-disjoint paths** is a set of paths, such that:
1. each of these paths must start in some  $s \in S$  and must end in  $t$ ; and
  2. none of these paths share an edge.

Let us look at an example:



Let  $S = \{s_1, s_2, s_3\}$ . Then the set consisting of the paths

$$s_1 \rightarrow v_1 \rightarrow t, \quad s_2 \rightarrow v_3 \rightarrow t, \quad s_3 \rightarrow v_2 \rightarrow t$$

is edge-disjoint. However, if we add the path  $s_1 \rightarrow v_2 \rightarrow t$ , then it is not edge-disjoint anymore.

In this exercise, we look the following problem: given  $G$ ,  $S$  and  $t$  as above, and for each element  $s \in S$  an natural number  $k_s \geq 0$ , is there a collection of edge disjoint paths such that for each  $s \in S$ , this collection contains at least  $k_s$  paths from  $s$  to  $t$ ? Describe an algorithm that solves this problem and returns a boolean that indicates where such a collection exists. Provide an analysis of the complexity and correctness of your algorithm.

4. (20 points) Consider a set of mobile computing *clients* in a certain town who each need to be connected to one of several possible *base stations*. There are  $n$  clients, with the position of each client specified by its  $(x, y)$  coordinates in the plane. There are also  $k$  base stations; the position of each of these is specified by  $(x, y)$  coordinates as well.

We wish to connect each client to exactly one of the base stations. Our choice of connection is constrained in the following ways. There is a *range parameter*  $r$  — a client can only be connected to a base station that is within distance  $r$ . There is also a *load parameter*  $L$  — no more than  $L$  clients can be connected to any single station.

Design a polynomial-time algorithm for the following problem. Given the positions of a set of clients and a set of base stations, as well as the range and load parameters, decide whether every client can be connected simultaneously to a base station, subject to the conditions from the previous paragraph.

5. (25 points) The Department of Computer Science at Radboud University introduces a flexible curriculum with a complex set of graduation requirements. The department offers  $n$  different courses, and there are  $m$  different requirements. Each requirement specifies a subset of the  $n$  courses and the number of courses that must be taken from that subset. The subsets for different requirements may overlap, but each course can only be used to satisfy at most one requirement.

For example, suppose there are  $n = 5$  courses  $A, B, C, D, E$  and  $m = 2$  requirements:

- You must take at least 2 courses from the subset  $\{A, B, C\}$ .
- You must take at least 2 courses from the subset  $\{C, D, E\}$ .

Then a student who has only taken courses  $B, C, D$  cannot graduate, but a student who has taken either  $A, B, C, D$  or  $B, C, D, E$  can graduate. Describe and analyze an efficient algorithm to determine whether a given student can graduate. The input to your algorithm is the list of  $m$  requirements (each specifying a subset of the  $n$  courses and the number of courses that must be taken from that subset) and the list of courses the student has taken.